PHY101 lesson wise solved mcqs for midterm by Mudasar Qureshi vu360

Lesson 1: Introduction to Physics

1. What is the definition of physics?

a) Study of living things

b) Study of non-living things

c) Study of matter and energy

d) Study of space and time

Answer: c) Study of matter and energy

1. Who is considered the father of physics?

a) Galileo Galilei

b) Isaac Newton

c) Albert Einstein

d) Aristotle

Answer: b) Isaac Newton

1. What is the SI unit of length?

a) Meter

b) Centimeter

c) Kilometer

d) Millimeter

Answer: a) Meter

1. What is the SI unit of mass?

a) Gram

b) Kilogram

c) Milligram

d) Ton

Answer: b) Kilogram

1. What is the SI unit of time?

a) Second

b) Minute

c) Hour

d) Day

Answer: a) Second

...and so on.

Lesson 2: Motion

1. What is the definition of motion?

a) Change of position with time

b) Change of velocity with time

c) Change of acceleration with time

d) Change of force with time

Answer: a) Change of position with time

1. What is the SI unit of velocity?

a) Meter per second

b) Kilometer per hour

c) Centimeter per second

d) Millimeter per second

Answer: a) Meter per second

1. What is the SI unit of acceleration?

a) Meter per second squared

b) Kilometer per hour squared

c) Centimeter per second squared

d) Millimeter per second squared

Answer: a) Meter per second squared

1. What is the equation of motion?

a) $v = u + at$

b) $s = ut + 1/2at^2$

c) $v^2 = u^2 + 2as$

d) $a = \Delta v / \Delta t$

Answer: b) $s = ut + 1/2at^2$

1. What is the concept of inertia?

a) Resistance to change in motion

b) Force required to change motion

c) Energy required to change motion

d) Momentum required to change motion

Answer: a) Resistance to change in motion

...and so on.

Lesson 3: Forces

1. What is the definition of force?

a) Push or pull

b) Energy or work

c) Motion or velocity

d) Acceleration or inertia

Answer: a) Push or pull

1. What is the SI unit of force?

a) Newton

b) Dyne

c) Joule

d) Kilogram

Answer: a) Newton

1. What is the equation for force?

a) F = ma

b) F = mv

c) F = ml

d) F = mt

Answer: a) $F = ma$

1. What is the concept of friction?

a) Force that opposes motion

b) Force that causes motion

c) Force that changes motion

d) Force that maintains motion

Answer: a) Force that opposes motion

1. What is the type of force that acts through physical contact?

a) Frictional force

b) Normal force

c) Tension force

d) Air resistance force

Answer: a) Frictional force

...and so on.

Lesson 4: Energy

1. What is the definition of energy?

a) Ability to do work

b) Ability to cause motion

c) Ability to change motion

d) Ability to maintain motion

Answer: a) Ability to do work

1. What is the SI unit of energy?

a) Joule

b) Newton

c) Kilogram

d) Meter

Answer: a) Joule

1. What is the equation for kinetic energy?

a) KE = 1/2mv^2

b) KE = mv^2

c) KE = 2mv^2

d) KE = mv

Answer: a) $KE = 1/2mv^2$

1. What is the type of energy that an object possesses due to its motion?

a) Kinetic energy

b) Potential energy

c) Thermal energy

d) Electrical energy

Answer: a) Kinetic energy

1. What is the law of conservation of energy?

a) Energy cannot be created or destroyed, only converted

b) Energy can be created or destroyed, but not converted

c) Energy can be converted, but not created or destroyed

d) Energy cannot be converted, created, or destroyed

Answer: a) Energy cannot be created or destroyed, only converted

...and so on.

Lesson 5: Momentum

1. What is the definition of momentum?

a) Product of mass and velocity

b) Product of force and velocity

c) Product of energy and velocity

d) Product of acceleration and velocity

Answer: a) Product of mass and velocity

1. What is the SI unit of momentum?

a) Kilogram-meter per second

b) Newton-second

c) Joule-second

d) Meter per second squared

Answer: a) Kilogram-meter per second

1. What is the equation for momentum?

a) p = mv

b) p = ma

c) p = ml

d) p = mt

Answer: a) p = mv

1. What is the concept of conservation of momentum?

a) Momentum cannot be created or destroyed, only converted

b) Momentum can be created or destroyed, but not converted

c) Momentum can be converted, but not created or destroyed

d) Momentum cannot be converted, created, or destroyed

Answer: a) Momentum cannot be created or destroyed, only converted

1. What is the type of collision in which momentum is conserved?

a) Elastic collision

b) Inelastic collision

c) Perfectly elastic collision

d) Perfectly inelastic collision

Answer: a) Elastic collision

...and so on.

Lesson 6: Rotational Motion

1. What is the definition of rotational motion?

a) Motion around a fixed axis

b) Motion in a straight line

c) Motion in a circular path

d) Motion in a elliptical path

Answer: a) Motion around a fixed axis

1. What is the SI unit of angular velocity?

a) Radian per second

b) Degree per second

c) Revolution per second

d) Meter per second squared

Answer: a) Radian per second

1. What is the equation for angular velocity?

a) $\omega = \Delta\theta / \Delta t$

b) $\omega = \Delta\theta / \Delta x$

c) $\omega = \Delta\theta / \Delta y$

d) $\omega = \Delta\theta / \Delta z$

Answer: a) $\omega = \Delta\theta / \Delta t$

1. What is the concept of torque?

a) Force that causes rotation

b) Force that opposes rotation

c) Force that maintains rotation

d) Force that changes rotation

Answer: a) Force that causes rotation

1. What is the SI unit of torque?

a) Newton-meter

b) Newton-second

c) Joule-second

d) Meter per second squared

Answer: a) Newton-meter

...and so on.

Lesson 7: Gravitation

1. What is the definition of gravitation?

a) Force that attracts two objects

b) Force that repels two objects

c) Force that maintains two objects

d) Force that changes two objects

Answer: a) Force that attracts two objects

1. What is the SI unit of gravitational force?

a) Newton

b) Dyne

c) Joule

d) Kilogram

Answer: a) Newton

1. What is the equation for gravitational force?

a) F = G * (m1 * m2) / r^2

b) F = G * (m1 + m2) / r^2

c) F = G * (m1 - m2) / r^2

d) F = G * (m1 * m2) / r

Answer: a) F = G * (m1 * m2) / r^2

1. What is the concept of gravitational potential energy?

a) Energy an object has due to its motion

b) Energy an object has due to its position

c) Energy an object has due to its mass

d) Energy an object has due to its velocity

Answer: b) Energy an object has due to its position

1. What is the equation for gravitational potential energy?

a) U = m * g * h

b) U = m * g / h

c) U = m / g * h

d) U = m / g / h

Answer: a) U = m * g * h

...and so on.

Lesson 8: Work and Energy

1. What is the definition of work?

a) Force applied over a distance

b) Force applied over time

c) Force applied over velocity

d) Force applied over acceleration

Answer: a) Force applied over a distance

1. What is the SI unit of work?

a) Joule

b) Newton

c) Kilogram

d) Meter

Answer: a) Joule

1. What is the equation for work?

a) W = F * s

b) W = F / s

c) W = F * t

d) W = F / t

Answer: a) W = F * s

1. What is the concept of kinetic energy?

a) Energy an object has due to its motion

b) Energy an object has due to its position

c) Energy an object has due to its mass

d) Energy an object has due to its velocity

Answer: a) Energy an object has due to its motion

1. What is the equation for kinetic energy?

a) KE = 1/2 * m * v^2

b) KE = 1/2 * m / v^2

c) KE = 1/2 * m * v

d) KE = 1/2 * m / v

Answer: a) KE = 1/2 * m * v^2

...and so on.

Lesson 9: Power

1. What is the definition of power?

a) Work done per unit time

b) Energy transferred per unit time

c) Force applied per unit distance

d) Velocity achieved per unit time

Answer: a) Work done per unit time

1. What is the SI unit of power?

a) Watt

b) Joule

c) Newton

d) Kilogram

Answer: a) Watt

1. What is the equation for power?

a) P = W / t

b) P = W * t

c) P = F * s

d) P = F / s

Answer: a) $P = W / t$

1. What is the concept of efficiency?

a) Ratio of output power to input power

b) Ratio of input power to output power

c) Ratio of work done to energy transferred

d) Ratio of energy transferred to work done

Answer: a) Ratio of output power to input power

1. What is the equation for efficiency?

a) $\eta = P_{output} / P_{input}$

b) $\eta = P_{input} / P_{output}$

c) $\eta = W_{output} / W_{input}$

d) $\eta = W_{input} / W_{output}$

Answer: a) $\eta = P_{output} / P_{input}$

...and so on.

Lesson 10: Oscillations

1. What is the definition of oscillation?

a) Motion that repeats itself

b) Motion that does not repeat itself

c) Motion that is constant

d) Motion that is changing

Answer: a) Motion that repeats itself

1. What is the SI unit of frequency?

a) Hertz

b) Second

c) Meter

d) Kilogram

Answer: a) Hertz

1. What is the equation for frequency?

a) $f = 1 / T$

b) $f = T / 1$

c) $f = \omega / 2\pi$

d) $f = 2\pi / \omega$

Answer: a) $f = 1 / T$

1. What is the concept of simple harmonic motion?

a) Motion that is oscillatory and periodic

b) Motion that is oscillatory but not periodic

c) Motion that is periodic but not oscillatory

d) Motion that is neither oscillatory nor periodic

Answer: a) Motion that is oscillatory and periodic

1. What is the equation for simple harmonic motion?

a) $x = A * \sin(\omega t + \phi)$

b) $x = A * \cos(\omega t + \phi)$

c) $x = A * \tan(\omega t + \phi)$

d) $x = A * \cot(\omega t + \phi)$

Answer: a) $x = A * \sin(\omega t + \phi)$

...and so on.

Lesson 11: Waves

1. What is the definition of a wave?

a) A disturbance that travels through a medium

b) A disturbance that travels through a vacuum

c) A disturbance that remains in one place

d) A disturbance that does not travel

Answer: a) A disturbance that travels through a medium

1. What is the SI unit of wave speed?

a) Meter per second

b) Kilometer per hour

c) Centimeter per second

d) Millimeter per second

Answer: a) Meter per second

1. What is the equation for wave speed?

a) $v = \lambda f$

b) $v = \lambda/f$

c) $v = f/\lambda$

d) $v = 2\pi f/\lambda$

Answer: a) $v = \lambda f$

1. What is the concept of wave superposition?

a) The combination of two or more waves

b) The separation of two or more waves

c) The reflection of a wave

d) The refraction of a wave

Answer: a) The combination of two or more waves

1. What is the result of wave superposition?

a) A wave with a larger amplitude

b) A wave with a smaller amplitude

c) A wave with the same amplitude

d) No wave at all

Answer: a) A wave with a larger amplitude

...and so on.

Lesson 12: Sound

1. What is the definition of sound?

a) A form of electromagnetic radiation

b) A form of mechanical radiation

c) A form of thermal radiation

d) A form of electrical radiation

Answer: b) A form of mechanical radiation

1. What is the SI unit of sound frequency?

a) Hertz

b) Second

c) Meter

d) Kilogram

Answer: a) Hertz

1. What is the equation for sound speed?

a) v = 331 + 0.6T

b) v = 331 - 0.6T

c) v = 331 + 0.6t

d) v = 331 - 0.6t

Answer: a) $v = 331 + 0.6T$

1. What is the concept of sound intensity?

a) The energy transferred per unit area per unit time

b) The energy transferred per unit area per unit frequency

c) The energy transferred per unit area per unit wavelength

d) The energy transferred per unit area per unit speed

Answer: a) The energy transferred per unit area per unit time

1. What is the equation for sound intensity?

a) $I = P / A$

b) $I = A / P$

c) $I = P * A$

d) $I = A * P$

Answer: a) $I = P / A$

...and so on.

Lesson 13: Thermodynamics

1. What is the definition of thermodynamics?

a) The study of heat and temperature

b) The study of work and energy

c) The study of motion and forces

d) The study of electricity and magnetism

Answer: a) The study of heat and temperature

1. What is the SI unit of temperature?

a) Kelvin

b) Celsius

c) Fahrenheit

d) Rankine

Answer: a) Kelvin

1. What is the equation for thermal expansion?

a) $\Delta L = \alpha * \Delta T$

b) $\Delta L = \alpha / \Delta T$

c) $\Delta L = \alpha * L * \Delta T$

d) $\Delta L = \alpha / L * \Delta T$

Answer: a) $\Delta L = \alpha * \Delta T$

1. What is the concept of heat capacity?

a) The amount of heat required to change the temperature of an object

b) The amount of work required to change the temperature of an object

c) The amount of energy required to change the temperature of an object

d) The amount of momentum required to change the temperature of an object

Answer: a) The amount of heat required to change the temperature of an object

1. What is the equation for heat capacity?

a) $C = \Delta Q / \Delta T$

b) $C = \Delta Q * \Delta T$

c) $C = \Delta Q / \Delta t$

d) $C = \Delta Q * \Delta t$

Answer: a) $C = \Delta Q / \Delta T$

...and so on.

Lesson 14: Electricity

1. What is the definition of electricity?

a) The study of electric charges and currents

b) The study of magnetic fields and forces

c) The study of thermal energy and temperature

d) The study of mechanical energy and motion

Answer: a) The study of electric charges and currents

1. What is the SI unit of electric charge?

a) Coulomb

b) Ampere

c) Volt

d) Ohm

Answer: a) Coulomb

1. What is the equation for electric force?

a) F = k * q1 * q2 / r^2

b) F = k * q1 / q2 * r^2

c) F = k * q1 * q2 / r

d) F = k * q1 / q2 * r

Answer: a) F = k * q1 * q2 / r^2

1. What is the concept of electric potential?

a) The potential energy per unit charge

b) The kinetic energy per unit charge

c) The thermal energy per unit charge

d) The mechanical energy per unit charge

Answer: a) The potential energy per unit charge

1. What is the equation for electric potential?

a) V = k * q / r

b) V = k * q * r

c) V = k / q * r

d) V = k / q / r

Answer: a) V = k * q / r

...and so on.

Lesson 15: Magnetism

1. What is the definition of magnetism?

a) The study of magnetic fields and forces

b) The study of electric charges and currents

c) The study of thermal energy and temperature

d) The study of mechanical energy and motion

Answer: a) The study of magnetic fields and forces

1. What is the SI unit of magnetic field strength?

a) Tesla

b) Gauss

c) Maxwell

d) Weber

Answer: a) Tesla

1. What is the equation for magnetic force?

a) F = qv x B

b) F = qv / B

c) F = qv * B

d) F = qv + B

Answer: a) $F = qv \times B$

1. What is the concept of magnetic induction?

a) The production of an electric current by a changing magnetic field

b) The production of a magnetic field by an electric current

c) The production of thermal energy by an electric current

d) The production of mechanical energy by an electric current

Answer: a) The production of an electric current by a changing magnetic field

1. What is the equation for magnetic induction?

a) $\varepsilon = -N * \Delta\Phi / \Delta t$

b) $\varepsilon = N * \Delta\Phi / \Delta t$

c) $\varepsilon = -N / \Delta\Phi * \Delta t$

d) $\varepsilon = N / \Delta\Phi * \Delta t$

Answer: a) $\varepsilon = -N * \Delta\Phi / \Delta t$

...and so on.

Lesson 16: Electromagnetic Induction

1. What is the definition of electromagnetic induction?

a) The production of an electric current by a changing magnetic field

b) The production of a magnetic field by an electric current

c) The production of thermal energy by an electric current

d) The production of mechanical energy by an electric current

Answer: a) The production of an electric current by a changing magnetic field

1. What is the equation for electromagnetic induction?

a) $\varepsilon = -N * \Delta\Phi / \Delta t$

b) $\varepsilon = N * \Delta\Phi / \Delta t$

c) $\varepsilon = -N / \Delta\Phi * \Delta t$

d) $\varepsilon = N / \Delta\Phi * \Delta t$

Answer: a) $\varepsilon = -N * \Delta\Phi / \Delta t$

1. What is the concept of inductance?

a) The ability of a conductor to store energy in a magnetic field

b) The ability of a conductor to store energy in an electric field

c) The ability of a conductor to store energy in a thermal field

d) The ability of a conductor to store energy in a mechanical field

Answer: a) The ability of a conductor to store energy in a magnetic field

1. What is the equation for inductance?

a) L = μ * N^2 * A / l

b) L = μ * N^2 / A * l

c) L = μ / N^2 * A * l

d) L = μ / N^2 / A / l

Answer: a) L = μ * N^2 * A / l

...and so on.

Lesson 17: Alternating Current

1. What is the definition of alternating current?

a) A current that flows in one direction only

b) A current that flows in both directions

c) A current that remains constant

d) A current that varies with time

Answer: b) A current that flows in both directions

1. What is the equation for alternating current?

a) I = I0 sin(ωt + ϕ)

b) $I = I0 \cos(\omega t + \phi)$

c) $I = I0 \tan(\omega t + \phi)$

d) $I = I0 \cot(\omega t + \phi)$

Answer: a) $I = I0 \sin(\omega t + \phi)$

1. What is the concept of impedance?

a) The total opposition to the flow of an alternating current

b) The total opposition to the flow of a direct current

c) The total opposition to the flow of a thermal current

d) The total opposition to the flow of a mechanical current

Answer: a) The total opposition to the flow of an alternating current

1. What is the equation for impedance?

a) $Z = \sqrt{}(R^2 + X^2)$

b) $Z = \sqrt{}(R^2 - X^2)$

c) $Z = \sqrt{}(R^2 + X)$

d) $Z = \sqrt{}(R^2 - X)$

Answer: a) $Z = \sqrt{}(R^2 + X^2)$

...and so on.

Lesson 18: Electromagnetic Waves

1. What is the definition of electromagnetic waves?

a) Waves that propagate through a medium

b) Waves that propagate through a vacuum

c) Waves that propagate through a magnetic field

d) Waves that propagate through an electric field

Answer: b) Waves that propagate through a vacuum

1. What is the equation for the speed of electromagnetic waves?

a) $c = 1 / \sqrt{(\mu 0 * \varepsilon 0)}$

b) $c = \sqrt{(\mu 0 * \varepsilon 0)}$

c) $c = \mu 0 * \varepsilon 0$

d) $c = 1 / (\mu 0 * \varepsilon 0)$

Answer: a) $c = 1 / \sqrt{(\mu 0 * \varepsilon 0)}$

1. What is the concept of radiation pressure?

a) The pressure exerted by electromagnetic waves on a surface

b) The pressure exerted by mechanical waves on a surface

c) The pressure exerted by thermal waves on a surface

d) The pressure exerted by electrical waves on a surface

Answer: a) The pressure exerted by electromagnetic waves on a surface

1. What is the equation for radiation pressure?

a) P = I / c

b) P = I * c

c) P = I / c^2

d) P = I * c^2

Answer: a) P = I / c

...and so on.

Lesson 19: Optics

1. What is the definition of optics?

a) The study of the behavior and properties of light

b) The study of the behavior and properties of sound

c) The study of the behavior and properties of thermal energy

d) The study of the behavior and properties of mechanical energy

Answer: a) The study of the behavior and properties of light

1. What is the equation for the refractive index?

a) n = c / v

b) $n = v / c$

c) $n = c * v$

d) $n = v / c^2$

Answer: a) $n = c / v$

1. What is the concept of total internal reflection?

a) The reflection of light at a surface

b) The refraction of light at a surface

c) The diffraction of light at a surface

d) The transmission of light at a surface

Answer: a) The reflection of light at a surface

1. What is the equation for the critical angle?

a) $\sin(\theta c) = n1 / n2$

b) $\sin(\theta c) = n2 / n1$

c) $\sin(\theta c) = 1 / n1$

d) $\sin(\theta c) = 1 / n2$

Answer: a) $\sin(\theta c) = n1 / n2$

...and so on.

Lesson 20: Modern Physics

1. What is the definition of modern physics?

a) The study of the behavior and properties of matter and energy at the atomic and subatomic level

b) The study of the behavior and properties of matter and energy at the macroscopic level

c) The study of the behavior and properties of thermal energy

d) The study of the behavior and properties of mechanical energy

Answer: a) The study of the behavior and properties of matter and energy at the atomic and subatomic level

1. What is the equation for the energy of a photon?

a) $E = hf$

b) $E = h / f$

c) $E = hc / \lambda$

d) $E = hc * \lambda$

Answer: a) $E = hf$

1. What is the concept of wave-particle duality?

a) The idea that particles can exhibit both wave-like and particle-like behavior

b) The idea that particles can only exhibit wave-like behavior

c) The idea that particles can only exhibit particle-like behavior

d) The idea that particles can neither exhibit wave-like nor particle-like behavior

Answer: a) The idea that particles can exhibit both wave-like and particle-like behavior

1. What is the equation for the de Broglie wavelength?

a) $\lambda = h / p$

b) $\lambda = h * p$

c) $\lambda = 1 / h * p$

d) $\lambda = 1 / h / p$

Answer: a) $\lambda = h / p$

...and so on.

Lesson 21: Nuclear Physics

1. What is the definition of nuclear physics?

a) The study of the behavior and properties of atomic nuclei

b) The study of the behavior and properties of electrons

c) The study of the behavior and properties of protons

d) The study of the behavior and properties of neutrons

Answer: a) The study of the behavior and properties of atomic nuclei

1. What is the equation for the binding energy of a nucleus?

a) BE = (Z * mp + N * mn) - m

b) BE = (Z * mp - N * mn) + m

c) BE = (Z * mp + N * mn) + m

d) BE = (Z * mp - N * mn) - m

Answer: a) BE = (Z * mp + N * mn) - m

1. What is the concept of nuclear stability?

a) The ability of a nucleus to resist changes in its structure

b) The ability of a nucleus to undergo changes in its structure

c) The ability of a nucleus to emit radiation

d) The ability of a nucleus to absorb radiation

Answer: a) The ability of a nucleus to resist changes in its structure

1. What is the equation for the half-life of a radioactive substance?

a) $t1/2 = \ln(2) / \lambda$

b) $t1/2 = \lambda / \ln(2)$

c) $t1/2 = 1 / \lambda$

d) $t1/2 = \lambda$

Answer: a) $t1/2 = \ln(2) / \lambda$

...and so on.

Lesson 22: Particle Physics

1. What is the definition of particle physics?

a) The study of the behavior and properties of subatomic particles

b) The study of the behavior and properties of atomic nuclei

c) The study of the behavior and properties of electrons

d) The study of the behavior and properties of protons

Answer: a) The study of the behavior and properties of subatomic particles

1. What is the equation for the energy of a particle in a accelerator?

a) E = γ * mc^2

b) E = 1 / γ * mc^2

c) E = γ / mc^2

d) E = 1 / γ / mc^2

Answer: a) E = γ * mc^2

1. What is the concept of conservation of momentum?

a) The idea that the total momentum of a closed system remains constant over time

b) The idea that the total energy of a closed system remains constant over time

c) The idea that the total angular momentum of a closed system remains constant over time

d) The idea that the total spin of a closed system remains constant over time

Answer: a) The idea that the total momentum of a closed system remains constant over time

1. What is the equation for the relativistic momentum of a particle?

a) p = γ * mu

b) p = 1 / γ * mu

c) p = γ / mu

d) p = 1 / γ / mu

Answer: a) p = γ * mu